

# User Guide

## for uCoupler

Revision: 1.1

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# 1. Overview

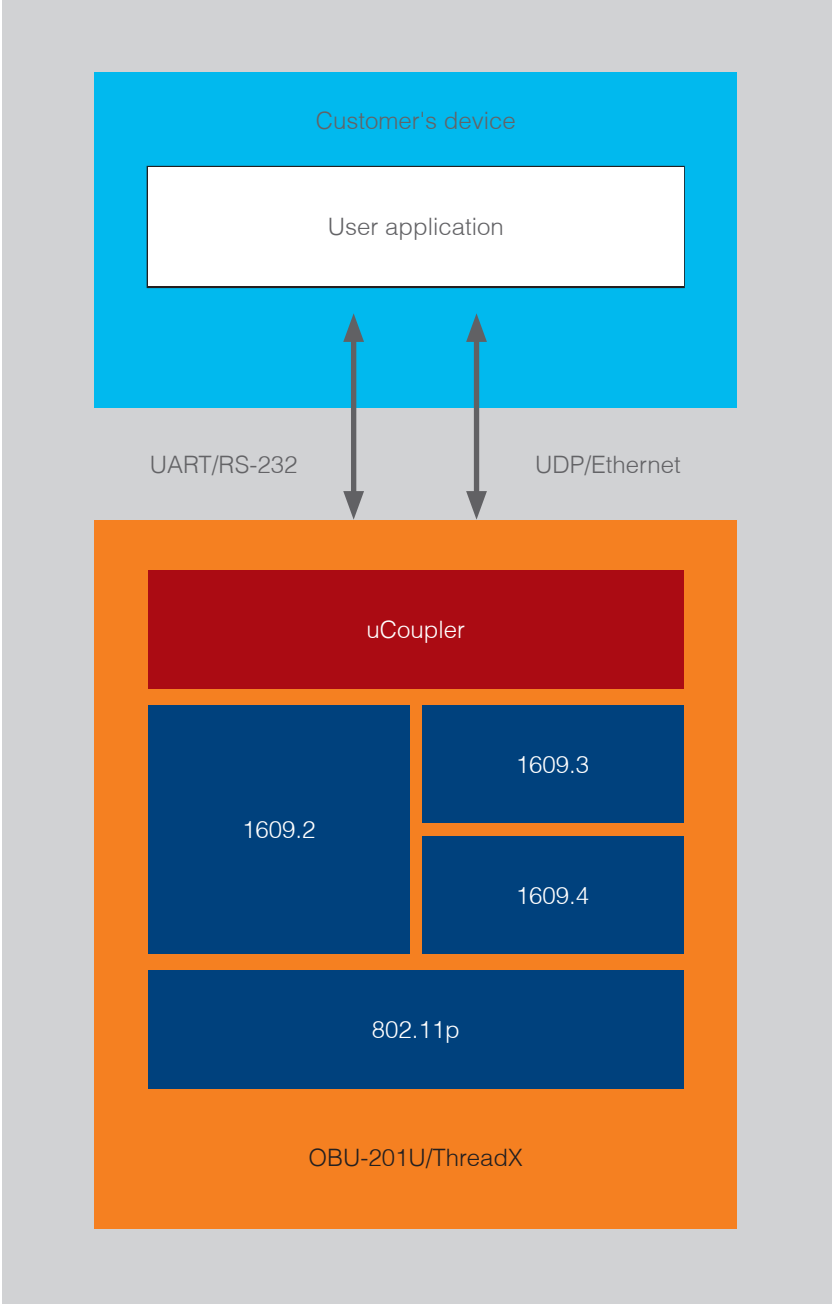



**Figure 1 - uCoupler System Overview**

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One of the vital use cases of OBU-201U (RSU-101U) is a message forwarder. With the built-in **uCoupler** middleware, the OBU-201U/RSU-101U can serve as a good solution for trials, retrofit or even C-ITS deployments. A user can run V2X applications (like SAE J2735 messages or safety warnings) on a PC/NB, an in-vehicle computer, or a traffic signal controller, then forward those messages to OBU-201U/RSU-101U via Ethernet or UART. OBU-201U/RSU-101U will send those messages in WSM messages over a 5.9GHz DSRC channel.

For this kind of use cases, customer's device behaves as "message generator/receiver", and OBU-201U/RSU-101U will act as "message forwarder" which forwards the "payload" part of received messages from a wire-connected device to another DSRC device. Figure 1 depicts an overall view of uCoupler while Figure 2 illustrates the mechanism.

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# 2. Mechanism

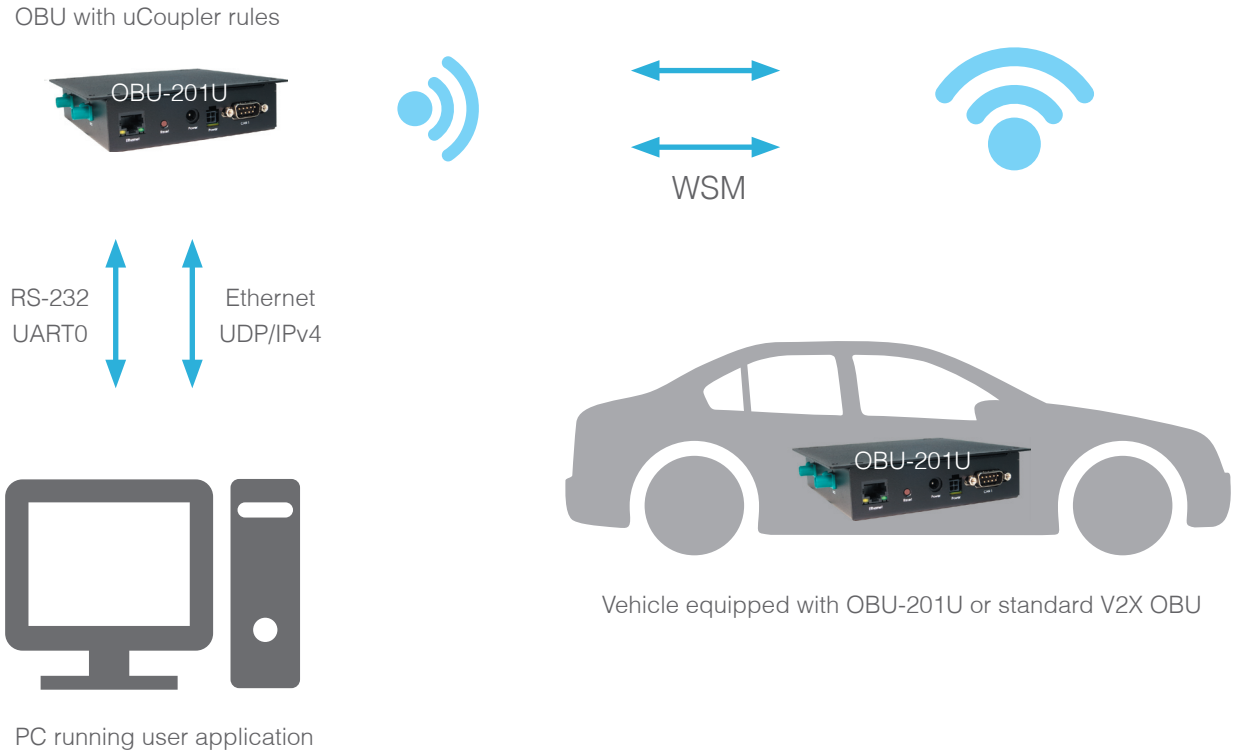



**Figure 2 - uCoupler Mechanism**

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# 3. Setting uCoupler Rules Using Configuration Files

To setup the conversion rule between PC and OBU-201U/RSU-101U, configuration files should be provided. User can modify PSID, tx power, data rate, radio number and channel number for different user applications. For building up an uCoupler service, a common PSID and the same radio channel are required for both the sender and receiver side, however the radio interface, data rate and tx power can be set different on each end.

When a UDP packet is received from the PC, the payload is extracted and inserted into a WSM packet with the user-specific PSID. Likewise, when a WSM packet is received with the same PSID, the payload will be extracted and inserted into a UDP packet and then sent to the PC of the receiver side. For data received on the UART interface, the payload is treated pretty much the same way as in above UDP process, but in UART frames instead of packets.

User can use an FTP client (such as FileZilla) to upload the configuration file to the NOR flash partition (B:/) of OBU-201U/RSU-101U. The FTP transfer type should be set as binary.

## Configuration file example:

```
#
# Configuration file for uCoupler service.
#
#======================= Global Settings =======================

[global]

# Channel number of the first DSRC interface
```

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```
   dsrc0_channel = 178

# Channel number of the second DSRC interface
   dsrc1_channel = 172

#======================= Service Settings ======================

# Up to 10 services can be added, but only 1 UART service allowed.
# [service_name]: name of the service assigned by user
# serv_type: IPV4 or UART
# serv_radio: 0 or 1. DSRC radio interface
# serv_psid: 0 to 270549119 (int). Use 16480 to 16511 for testing
# data_rate: 6, 9, 12, 18, 24, 36, 48 and 54 (500kbps)
# tx_power: 1 to 20 (int, dBm)
# remote_ipv4_addr: IP address of the device connected via Ethernet
# remote_ipv4_port: UDP port of the device connected via Ethernet

[test_wsm]
   serv_type = IPV4
   serv_radio = 0
   serv_psid = 16480
   data_rate = 6
   tx_power = 20
   remote_ipv4_addr = 192.168.1.10
   remote_ipv4_port = 4444

[test_channel_1]
   serv_type = IPV4
   serv_radio = 1
   serv_psid = 16481
   data_rate = 6
   tx_power = 20
   remote_ipv4_addr = 192.168.10.10
```

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```
    remote_ipv4_port = 4445


[test_uart]
  serv_type = UART
  serv_radio = 0
  serv_psid = 16482
  data_rate = 18
  tx_power = 16
```

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# 4. Example Testing

## 4.1. Ethernet-to-uCoupler Functionality

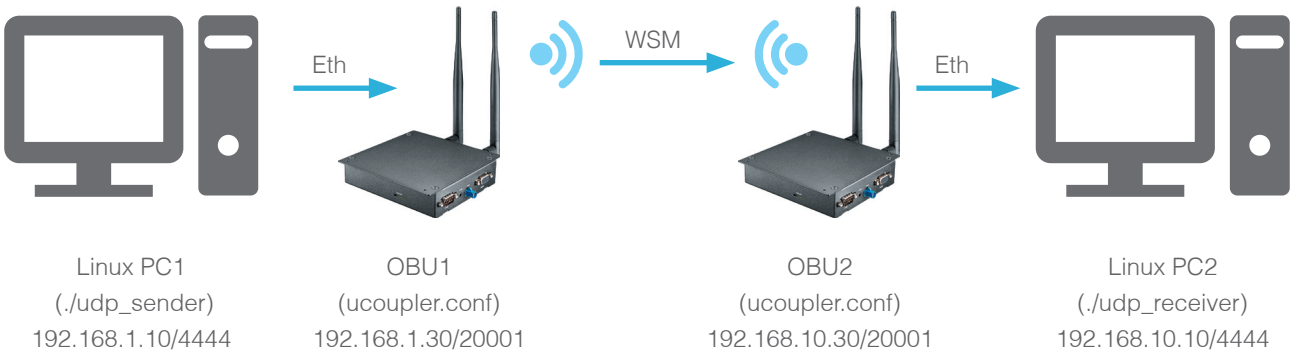



**Figure 3 - Ethernet-to-uCoupler Environment**

### Pre-requisites:

1. Configuration files for 2 OBU-201U units (sender & receiver) are provided.
2. A customized FW (wave-sdk-ucoupler.img) is provided.
3. Example codes to be executed on PC (Linux) are also provided.

### Limitations:

1. User cannot assign the same PSID to multiple services.
2. Only WSM packets can be transmitted on DSRC interfaces. It means users can not send/receive IP packets on DSRC interfaces when using uCoupler.

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## Configuration File Example:

OBU1 (only Service Settings listed):

```
[test_wsm]
   serv_type = IPV4
   serv_radio = 0
   serv_psid = 16480
   data_rate = 6
   tx_power = 20
   remote_ipv4_addr = 192.168.1.10
   remote_ipv4_port = 4444
```

OBU2 (only Service Settings listed):

```
[test_wsm]
   serv_type = IPV4
   serv_radio = 0
   serv_psid = 16480
   data_rate = 6
   tx_power = 20
   remote_ipv4_addr = 192.168.10.10
   remote_ipv4_port = 4444
```

## Procedure:

**Step 1:** Upload the configuration file over FTP via an FTP client (such as FileZilla) onto the NOR Flash partition (B:/) of OBU-201U on each end of the forwarder pair respectively. The FTP transfer type should be set as binary. Note that the IP address of OBU2 is different from

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OBU1, hence the `remote_ipv4_addr` in configuration files should be modified as such.

**Step 2:** Flash the FW (wave-sdk-ucoupler.img) onto the two OBU-201U units. Firmware upgrade procedure is available in Appendix. After flashing the FW, the following message will be displayed on CLI: `[UC] uCoupler firmware started` (as highlighted in Figure 4). The CLI of console will also be suspended. After this point onwards, user can only access the CLI via Telnet.

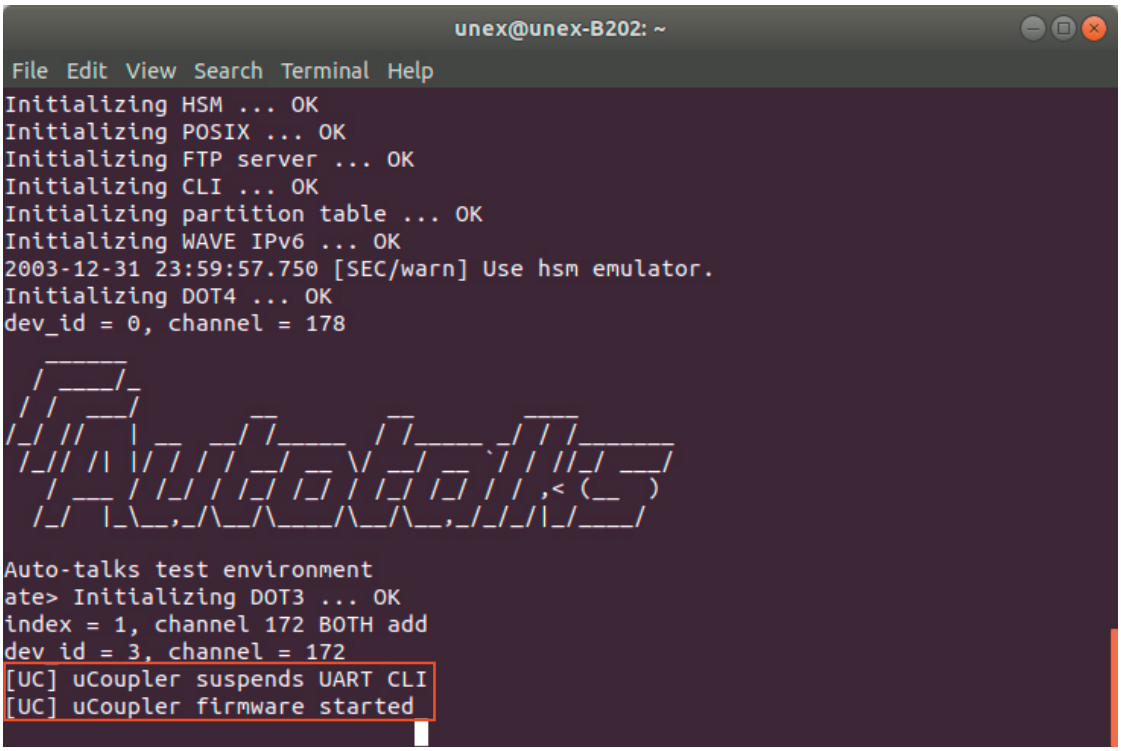



**Figure 4 - Snapshot After Firmware Upgrade**

**Step 3:** Compile the example code using gcc on Linux environment

**Example:** `./compile.sh`

**Note:** The source codes are available in the path `example/UDP/`. The pre-existing `udp_sender` and `udp_receiver` executables are compiled under 64-bit Linux environment.

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**Step 4:** Run the UDP sender executable on Linux PC1

**Usage:** `./udp_sender <remote_ip> <remote_port> <local_port> <pkt_num> <priod_ms> <pkt_size>`

**Example:** `./udp_sender 192.168.1.30 20001 4444 100 10 100`

(see Figure 5)

**Note:**

» `<remote_ip>`: OBU-201U IP address. User should match it with the IP address setting of OBU-201U.

» `<remote_port>`: OBU-201U port number. OBU-201U will always use port number 20001 as the sending/receiving port.

» `<local_port>`: PC port number. It should match with the `remote_ipv4_port` in the configuration file. User can use different UDP source ports on the PC end for different services.

» `<pkt_num>`: total count of packets transmitting via uCoupler service.

» `<period_ms>`: the period in milliseconds between two packets transmission.

» `<pkt_size>`: packet size being transmitted.

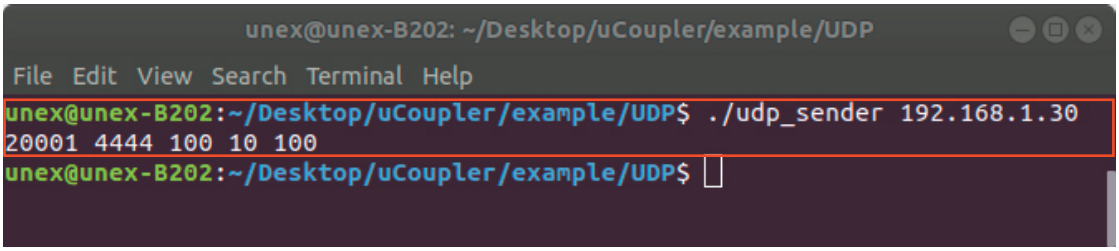



**Figure 5 - Snapshot of udp_sender Command**

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**Step 5:** Run the UDP receiver executable on Linux PC2

**Usage:** `./udp_receiver <local_port>`

**Example:** `./udp_receiver 4444`

(See Figure 6)

**Note:** <local_port>: PC port number. It should match with the `remote_ipv4_port` in the configuration file.

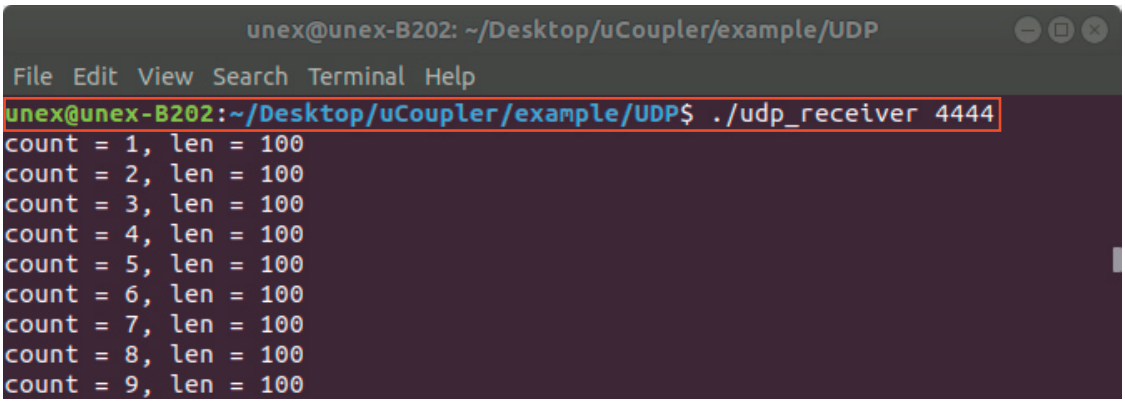



**Figure 6 - Snapshot of udp_receiver Command**

**Step 6:** Access the sender-side OBU-201U via Telnet and execute the following command. The total DSRC packets/bytes sent via the specific uCoupler service will be shown. (highlighted in Figure 7)

```
ate> uCoupler
```

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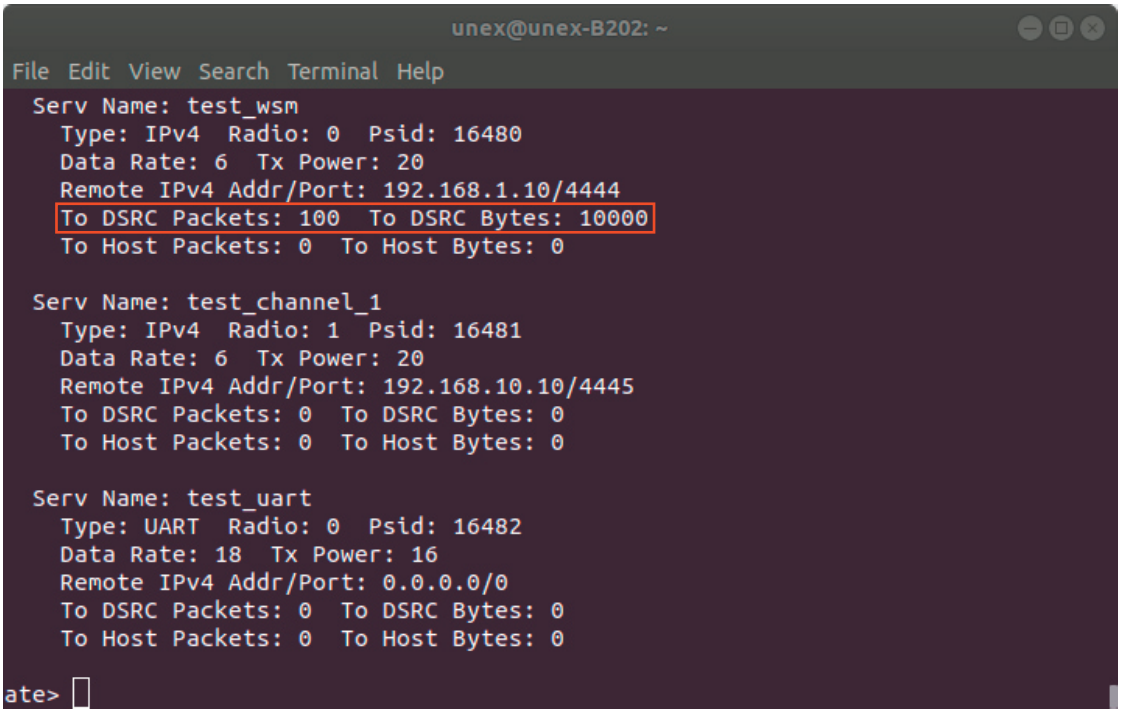



**Figure 7 - Snapshot of uCoupler at Sender-side**

**Step 7:** Similarly, access the receiver-side OBU-201U via Telnet and execute the following command. The total DSRC packets/bytes received via the specific uCoupler service will be shown. (highlighted in Figure 8)

```
ate> uCoupler
```

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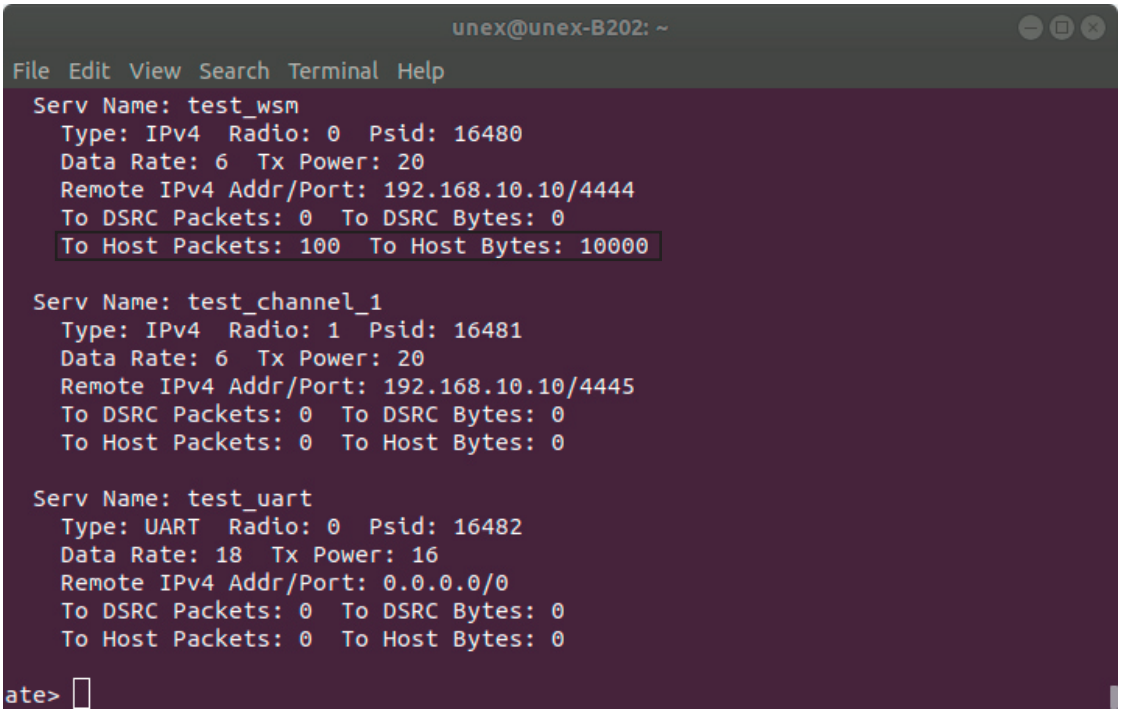



**Figure 8 - Snapshot of uCoupler at Receiver-side**

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# 4.2. UART-to-uCoupler Functionality

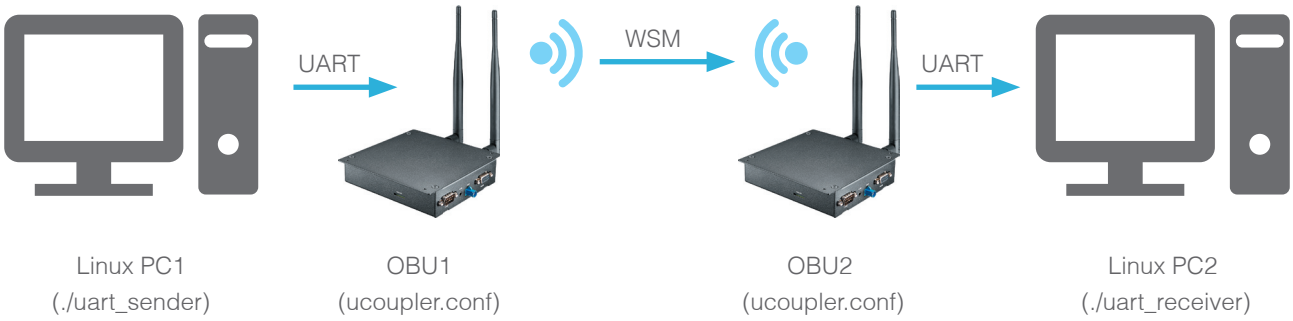



**Figure 9 - UDP-to-uCoupler Environment**

## Pre-requisites:

1. Configuration files for 2 OBU-201U units (sender & receiver) are provided.
2. A customized FW (wave-sdk-ucoupler.img) is provided.
3. Example codes to be executed on PC (Linux) are also provided.

## Limitations:

1. Console access will be disabled since the UART port of OBU-201U is used to send/receive data frames. User can only access the CLI via Telnet.
2. The maximum payload size of each UART frame is 1460 bytes.
3. User cannot assign the same PSID to multiple services.

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## Configuration File Example:

OBU1/OBU2 (only Service Settings listed):

```
[test_uart]
  serv_type = UART
  serv_radio = 0
  serv_psid = 16482
  data_rate = 18
  tx_power = 16
```

The same configuration file is used for both sender and receiver side. On OBU-201U, a UART service will be created. For a frame received on the UART interface, the payload is extracted and inserted into a WSM packet with user assigned PSID. Likewise, for a WSM packet received with the same PSID, the payload will be extracted and inserted into a UART frame and then sent to the UART receiver device.

## Procedure:

**Step 1:** Upload the configuration file over FTP via an FTP client (such as FileZilla) onto the NOR Flash partition (B:/) of OBU-201U on each end of the forwarder pair respectively. The FTP transfer type should be set as binary.

**Step 2:** Flash the FW (wave-sdk-ucoupler.img) onto the two OBU-201U units. Firmware upgrade procedure is available in Appendix. After flashing the FW, the following message will be displayed on CLI: `[UC] uCoupler firmware started` (as highlighted in Figure 10). The CLI of console will also be suspended. After this point onwards, user can only access the CLI via Telnet.

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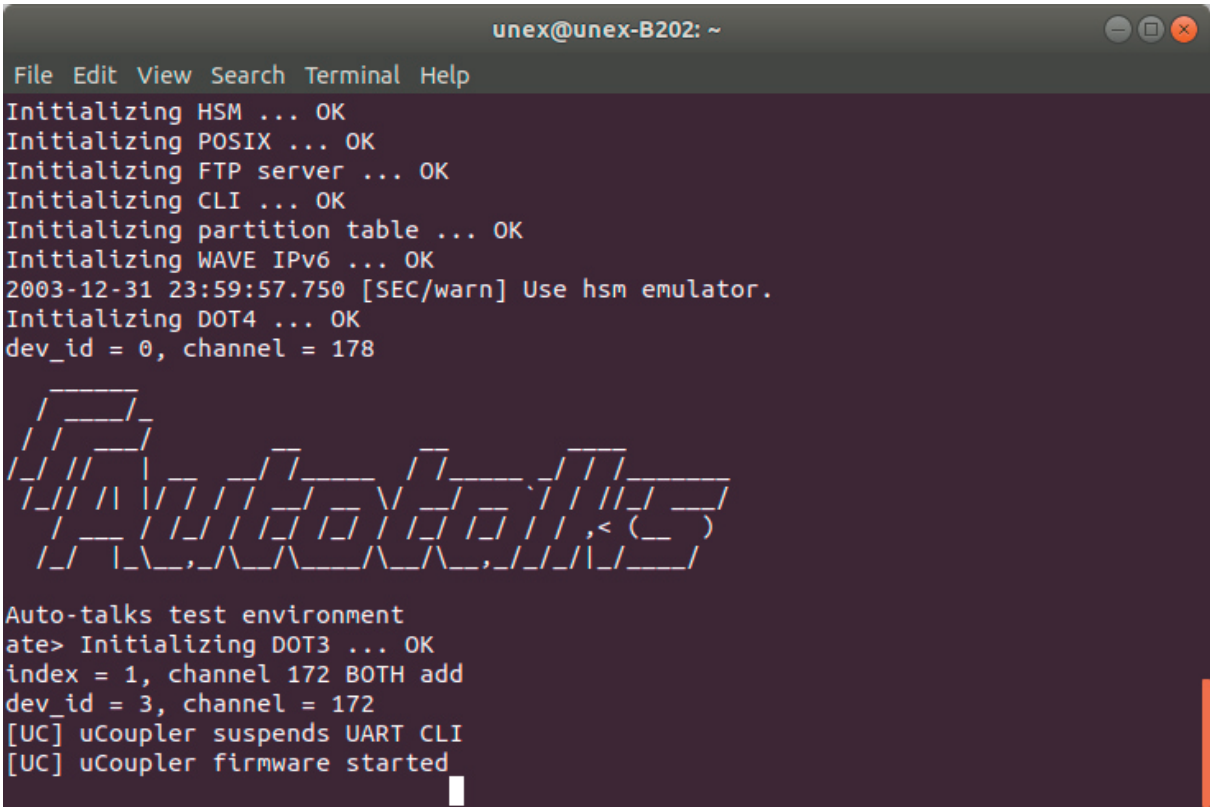



**Figure 10 - Snapshot After Firmware Upgrade**

**Step 3:** Compile the example code using gcc on Linux environment.

**Example:** `./compile.sh`

**Note:** The source codes are available in the path `example/UART/`. The pre-existing `uart_sender/uart_receiver` executables are compiled under 64-bit Linux environment.

**Step 4:** Run the UART sender executable on Linux PC1

**Usage:** `./uart_sender <dev_name> <pkt_num> <priod_ms> <pkt_size>`

**Example:** `sudo ./uart_sender /dev/ttyUSB0 20 10 100`

(See Figure 11)

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**Note:**

» `<dev_name>`: device name of the UART port on the PC. For a PC equipped with a USB-to-RS232 cable, /dev/ttyUSB0 should be the most common name.

» `<pkt_num>`: total count of packets transmitting via uCoupler service.

» `<period_ms>`: the period in milliseconds between two packets transmission.

» `<pkt_size>`: packet size being transmitted.

» sudo is needed for a non-root user to get access to that UART device.

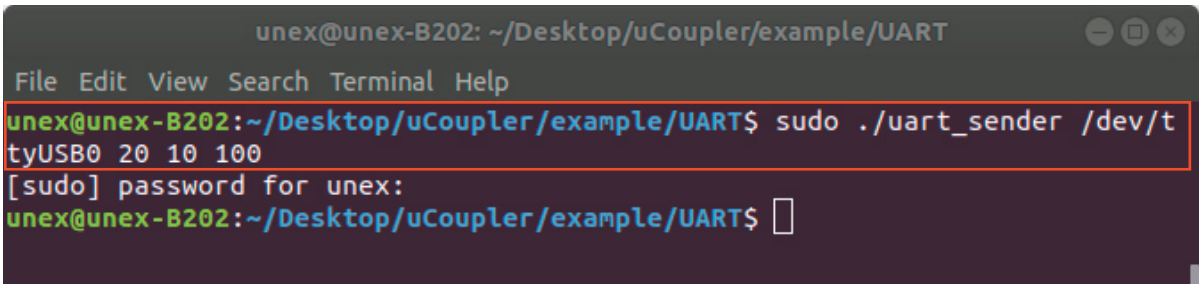



**Figure 11 - Snapshot of uart_sender Command**

**Step 5:** Run the UART receiver executable on Linux PC2

**Usage:** `./uart_receiver <dev_name>`

**Example:** `sudo ./uart_receiver /dev/ttyUSB0`

(See Figure 12)

**Note:**

» `<dev_name>`: device name of the UART port on the PC. For a PC equipped with a USB-to-RS232 cable, /dev/ttyUSB0 should be the most common name.

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» `sudo` is needed for a non-root user to get access to that UART device.

**Important Tip:**

Please terminate any other processes accessing the same UART port (e.g. minicom, putty, cu, screen, tip, etc.) before launching the uart_receiver. Accessing the same UART port with 2 processes will lead to serious data drop during transmission.

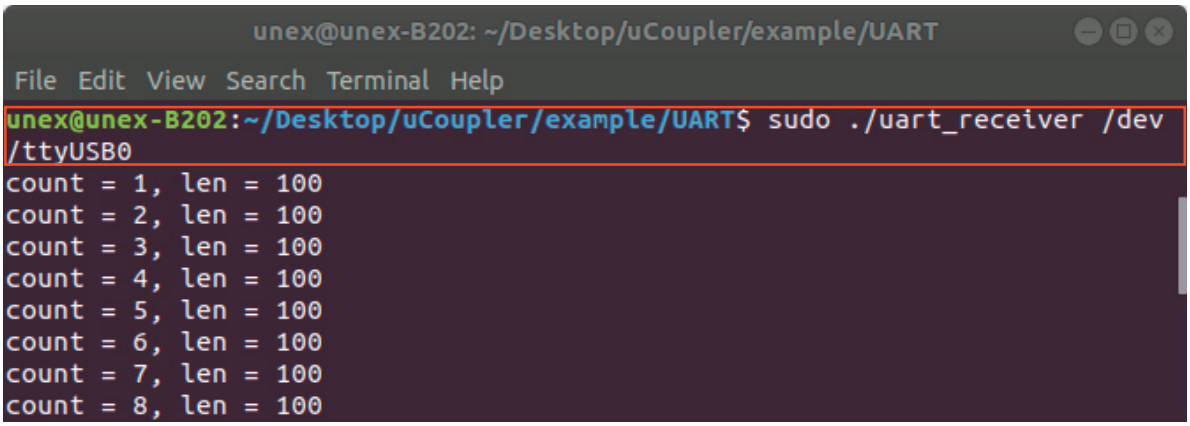



**Figure 12 - Snapshot of uart_receiver Command**

**Step 6:** Access the sender-side OBU-201U via Telnet and execute the following command. The total DSRC packets/bytes sent via the specific uCoupler service will be shown. (highlighted in Figure 13)

```
ate> uCoupler
```

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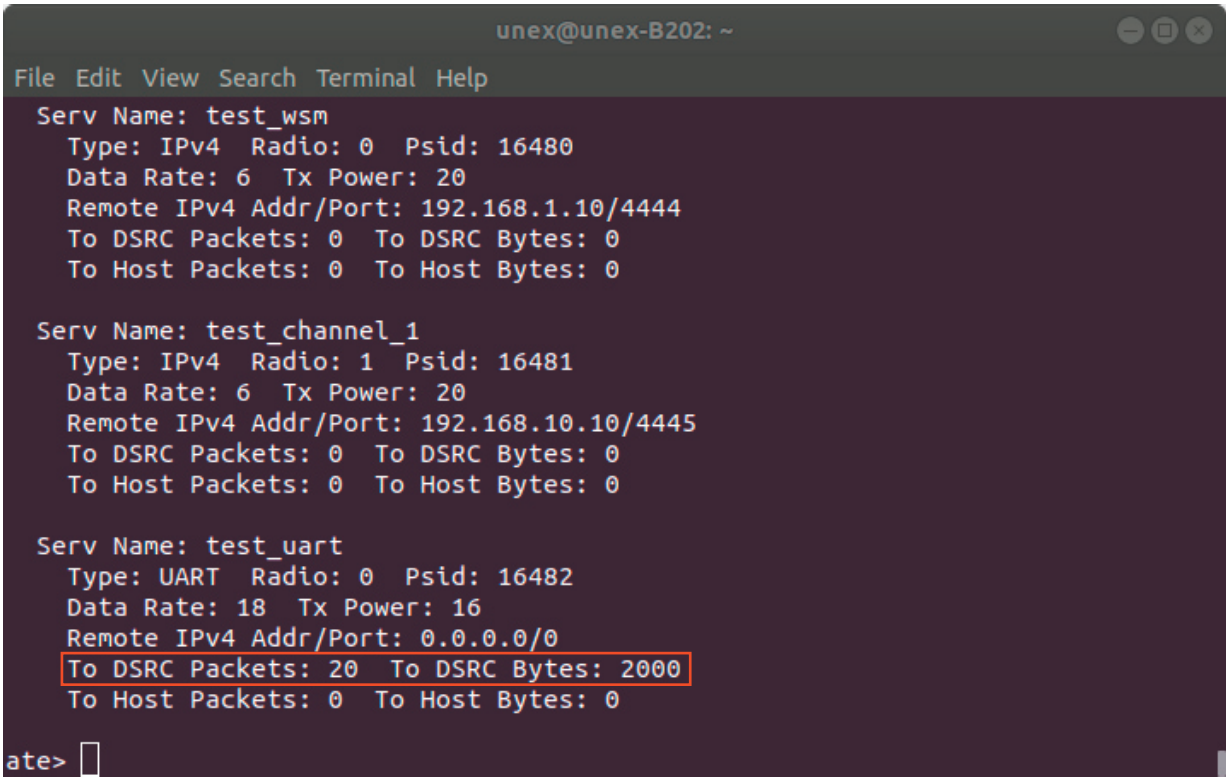



**Figure 13 - Snapshot of uCoupler at Sender-side**

**Step 7:** Similarly, access the receiver-side OBU-201U via Telnet and execute the following command. The total DSRC packets/bytes received via the specific uCoupler service will be shown. (highlighted in Figure 14)

```
ate> uCoupler
```

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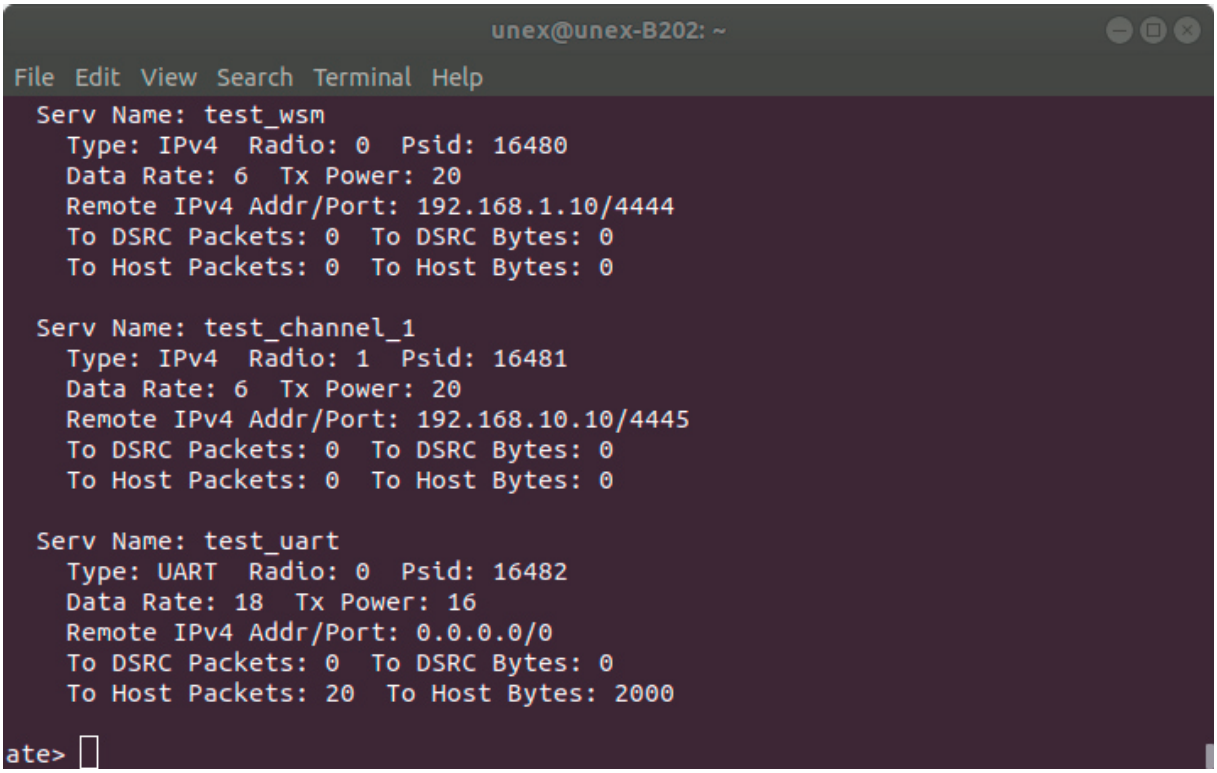



**Figure 14 - Snapshot of uCoupler at Receiver-side**

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# 4.3. Ethernet-to-UART Functionality

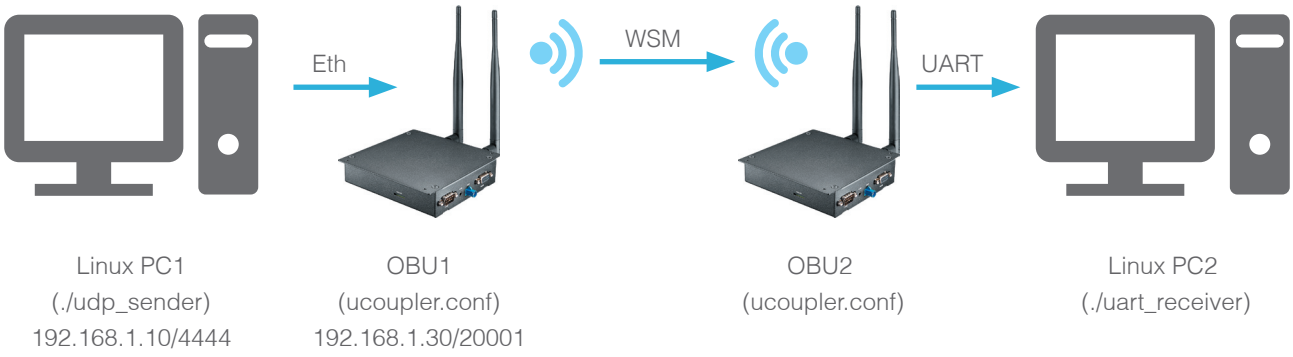



**Figure 15 - Ethernet-to-UART Environment**

## Pre-requisites:

1. Configuration files for 2 OBU-201U units (sender & receiver) are provided.
2. A customized FW (wave-sdk-ucoupler.img) is provided.
3. Example codes to be executed on PC (Linux) are also provided.

## Limitations:

1. User cannot assign the same PSID to multiple services.
2. Only WSM packets can be transmitted on DSRC interfaces. It means users can not send/receive IP packets on DSRC interfaces when using uCoupler.

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## Configuration File Example:

OBU1 (only Service Settings listed):

```
[test_wsm]
    serv_type = IPV4
    serv_radio = 0
    serv_psid = 16480
    data_rate = 6
    tx_power = 20
    remote_ipv4_addr = 192.168.1.10
    remote_ipv4_port = 4444
```

OBU2 (only Service Settings listed):

```
[test_uart]
   serv_type = UART
   serv_radio = 0
   serv_psid = 16480
   data_rate = 18
   tx_power = 16
```

## Procedure:

**Step 1:** Upload the configuration file over FTP via an FTP client (such as FileZilla) onto the NOR Flash partition (B:/) of OBU-201U on each end of the forwarder pair respectively. The FTP transfer type should be set as binary.

**Step 2:** Flash the FW (wave-sdk-ucoupler.img) onto the two OBU-201U units.

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Firmware upgrade procedure is available in Appendix. After flashing the FW, the following message will be displayed on CLI: `[UC] uCoupler firmware started` (as highlighted in Figure 16). The CLI of console will also be suspended. After this point onwards, user can only access the CLI via Telnet.

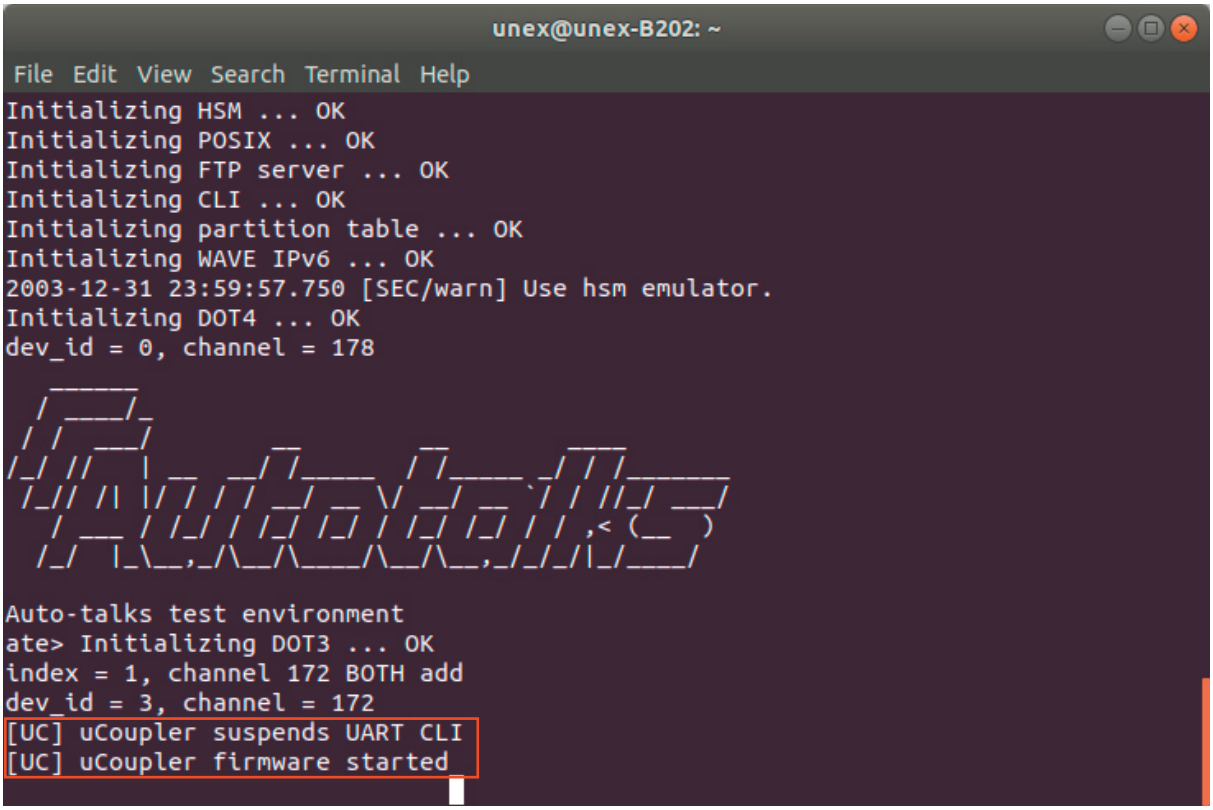



**Figure 16 - Snapshot After Firmware Upgrade**

**Step 3:** Compile the example code using gcc on Linux environment

**Example:** `./compile.sh`

**Note:** The source codes are available in the path `example/UDP/` and `example/UART/`. The pre-existing `udp_sender/uart_receiver` executables are compiled under 64-bit Linux environment.

**Step 4:** Run the UDP sender executable on Linux PC1

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**Usage:** `./udp_sender <remote_ip> <remote_port> <local_port> <pkt_num> <priod_ms> <pkt_size>`

**Example:** `./udp_sender 192.168.1.30 20001 4444 100 10 100`

(See Figure 17)

**Note:**

» `<remote_ip>`: OBU-201U IP address. User should match it with the IP address setting of OBU-201U.

» `<remote_port>`: OBU-201U port number. OBU-201U will always use port number 20001 as the sending/receiving port.

» `<local_port>`: PC port number. It should match with the `remote_ipv4_port` in the configuration file. User can use different UDP source ports on the PC end for different services.

» `<pkt_num>`: total count of packets transmitting via uCoupler service.

» `<period_ms>`: the period in milliseconds between two packets transmission.

» `<pkt_size>`: packet size being transmitted.

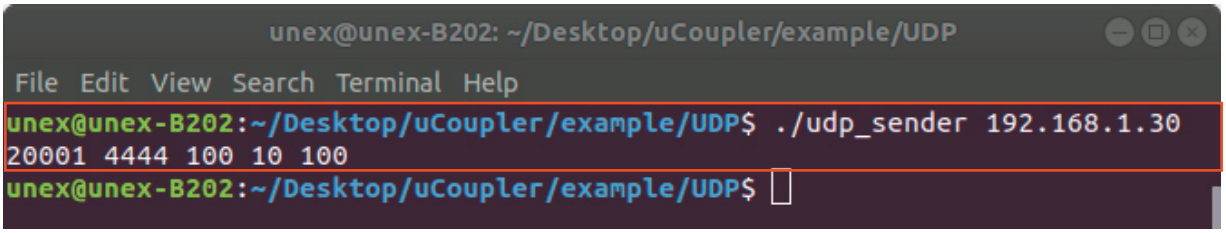



**Figure 17 - Snapshot of udp_sender Command**

**Step 5:** Run the UART receiver executable on Linux PC2

**Usage:** `./uart_receiver <dev_name>`

**Example:** `sudo ./uart_receiver /dev/ttyUSB0`

(See Figure 18)

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**Note:**

» `<dev_name>`: device name of the UART port on the PC. For a PC equipped with a USB-to-RS232 cable, /dev/ttyUSB0 should be the most common name.

» `sudo` is needed for a non-root user to get access to that UART device.

**Important Tip:**

Please terminate any other processes accessing the same UART port (e.g. minicom, putty, cu, screen, tip, etc.) before launching the uart_receiver. Accessing the same UART port with 2 processes will lead to serious data drop during transmission.

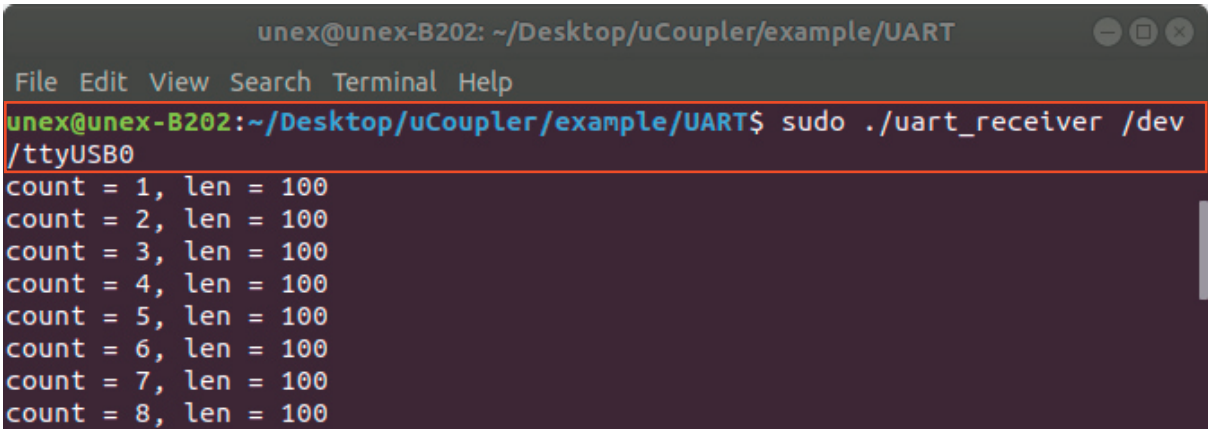



**Figure 18 - Snapshot of uart_receiver Command**

**Step 6:** Access the sender-side OBU-201U via Telnet and execute the following command. The total DSRC packets/bytes sent via the specific uCoupler service will be shown. (highlighted in Figure 19)

```
ate> uCoupler
```

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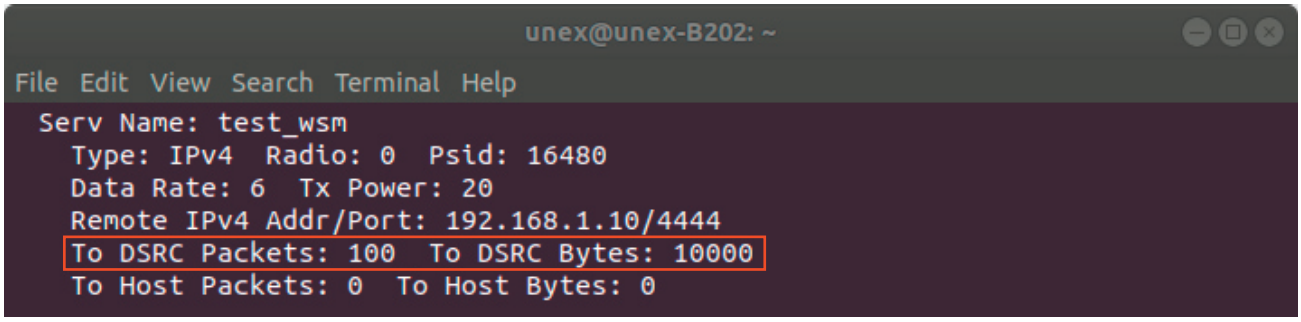



**Figure 19 - Snapshot of uCoupler at Sender-side**

**Step 7:** Similarly, access the receiver-side OBU-201U via Telnet and execute the following command. The total DSRC packets/bytes received via the specific uCoupler service will be shown. (highlighted in below snapshot)

```
ate> uCoupler
```

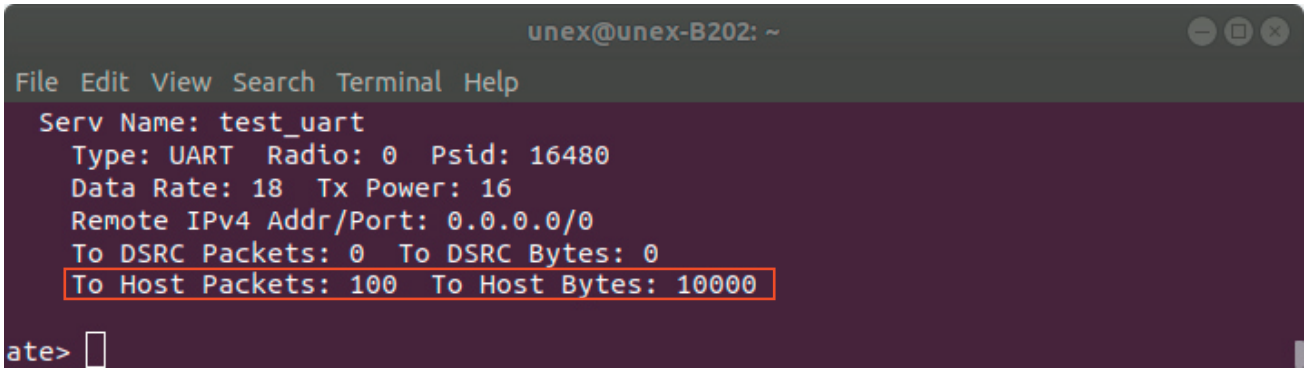



**Figure 20 - Snapshot of uCoupler at Receiver-side**

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# Appendix A: Firmware Upgrade Procedure

1. Load the firmware binary (wave-sdk-ucoupler.img) to the RAM partition (D:/) of OBU-201U/RSU-101U over FTP via a FTP client (such as FileZilla).
2. To flash the firmware, access the OBU-201U/RSU-101U via Telnet and execute the following command.

   ```
   ate> firmware update D:/wave-sdk-ucoupler.img
   ```

   **Note:** This step may consume about 40 seconds.
3. Reboot device.

   ```
   ate> reboot
   ```

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Unex Technology Corporation

11F-3, No. 100, Sec. 1, Jiafeng 11th Rd., Zhubei City,

Hsinchu County 302, Taiwan, R.O.C.

Tel. 886-3-657-8188

Fax. 886-3-657-8199

www.unex.com.tw